

AF-LeWM-lite on Official PushT

Matched-Compute Evaluation Summary

April 10, 2026

Bottom line. On official PushT, the original AF-LeWM-lite v2 objective remains the only AF variant that beats baseline control under the matched-compute budget, but the margin is small. The first 50-episode comparison gives **3/50** successes for v2 versus **2/50** for baseline, and a second 50-episode evaluation seed reduces the aggregate to **5/100** versus **4/100**. Follow-up sweeps show that the dynamics nuisance adversary creates the control gain and also causes the main degradation in the shared JEPA objective.

Setup

The study uses the official `pusht_expert_train.h5` extracted from `pusht_expert_train.h5.zst`. The active file is **46.30 GB** with **2,336,736** cached PushT samples. All primary models and follow-up sweep variants were trained on the same RTX 4070 Laptop GPU with the same fixed budget:

- 10 epochs
- 200 training batches per epoch
- 20 validation batches per epoch
- batch size 4
- bf16 training
- final 50-episode planning evaluation at epoch 10

This budget gives a fair relative comparison on the official dataset without turning the experiment into an impractically long full-data sweep.

Results

Primary matched-compute comparison

Model	Params (M)	Train (min)	Val pred	Core loss	Success (%)
Baseline LeWM	18.03	7.78	0.0035	0.1469	4.0
AF-LeWM-lite v1	18.17	12.51	0.0157	0.1649	2.0
AF-LeWM-lite v2	18.24	13.42	0.2538	0.4726	6.0

The parameter differences are small. AF-LeWM-lite v1 adds **0.74%** over baseline, and v2 adds **1.12%**. The planning ranking therefore is not explained by a large capacity jump.

The **core loss** is the shared JEPA objective,

$$\mathcal{L}_{\text{core}} = \mathcal{L}_{\text{pred}} + 0.09 \mathcal{L}_{\text{sigreg}},$$

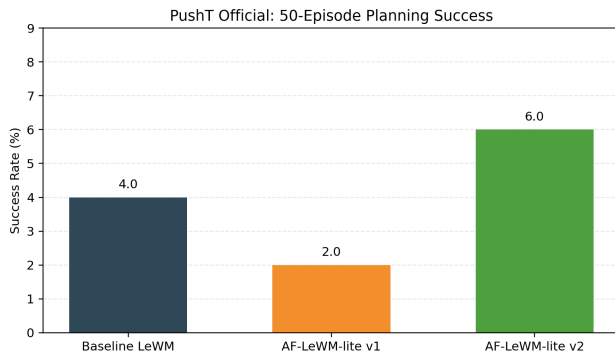
which is directly comparable across baseline, v1, and v2. The larger `validate/loss_epoch` used during v2 training is not comparable because it also includes the AF-specific nuisance regularizers.

Follow-up v2 sweep

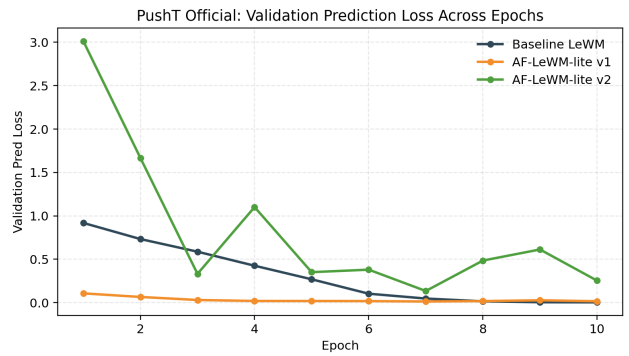
Model	Core loss	Success (%)	Comment
Baseline LeWM	0.1469	4.0	reference
AF-LeWM-lite v2	0.4726	6.0	original best 50-episode control
v2 no adversary	0.1491	0.0	removed dynamics nuisance adversary
v2 soft	0.3603	2.0	reduced all auxiliary weights
v2 GRL half	0.3145	2.0	kept weights, halved GRL lambda

Second-seed check

Model	Seed 42	Seed 43	Aggregate
Baseline LeWM	2/50	2/50	4/100 = 4.0%
AF-LeWM-lite v2	3/50	2/50	5/100 = 5.0%



(a) Final planning success on the 50-episode evaluation.



(b) Validation prediction loss across the 10 training epochs.

Interpretation

AF-LeWM-lite v1 is not a good stopping point. The split plus invariance and independence losses alone degrade both latent prediction quality and planning success on official PushT. This matches the earlier concern that `app_emb` lacked a strong positive job.

The dynamics nuisance adversary is the key mechanism. Removing it restores the shared core loss almost exactly to baseline, then collapses control to **0/50**. Weakening the adversarial pressure also pulls the core loss back down and reduces success to **1/50**. That means the adversary is not incidental regularization. It is the main source of the current control gain.

The tradeoff is real and currently sharp. Original v2 reaches the best control result while showing the worst shared core loss by a large margin. The follow-up sweep shows that static weight reduction does not solve this cleanly. Better shared modeling and better PushT control are still pulling in different directions under the current setup.

The control edge is fragile. The second 50-episode evaluation seed reduces the primary comparison to **5/100** for v2 versus **4/100** for baseline. That keeps v2 slightly ahead and removes any basis for a strong robustness claim.

Conclusion

The current conclusion is specific and actionable:

- Keep the **original AF-LeWM-lite v2** mechanism as the active branch.
- Drop **AF-LeWM-lite v1** as a candidate baseline for future reporting.
- Stop prioritizing static weight reduction. The follow-up sweep shows that weaker adversarial pressure improves the shared core loss and weakens control.
- Run the next experiment on **adversary scheduling**: delayed onset or gradual ramp of the dynamics nuisance adversary.
- Treat the current result as a **matched-compute PushT tradeoff study**, not a strong absolute PushT benchmark. All success rates remain low under this short budget.
- Use **both** shared core loss and success rate in future comparisons. Success alone is too noisy at 50 episodes, and total training loss is not cross-model comparable once AF-specific regularizers are active.
- Report multi-seed aggregates before claiming that AF-LeWM-lite improves PushT planning.

In one sentence: **the factorized-latent idea is still viable, the original v2 objective has a fragile control edge, and the highest-value next step is adversary schedule tuning rather than static loss reweighting or more model complexity.**

Artifacts generated in report/: `pusht_official_budget_summary.csv`, `pusht_official_budget_results.json`, `pusht_v2_followup_summary.csv`, `pusht_success_rate.png`, and `pusht_val_pred_loss.png`.